

STAT 654

Model Selection and Regularization

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Need for Regularization

- Consider the linear model: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$.
- Prediction Accuracy
 - If $n < p$ (no unique LS estimate) or n is not much larger than p (high variability in LS estimates).
 - By *constraining* or *shrinking* the estimated coefficients, we can reduce the variance at the cost of a negligible increase in bias.
 - Leads to substantial improvements in the accuracy with which we can predict the response for observations not used in model training.

Need for Regularization

- Model Interpretability
 - Some or many of the variables used in a multiple regression are not associated with the response.
 - By setting the corresponding coefficient estimates to 0 - we can obtain a model that is more easily interpreted.
 - LS is extremely unlikely to yield any coefficient estimates that are exactly 0.
 - Need approaches for automatically performing *feature selection* or *variable selection*, i.e., for excluding irrelevant variables from a multiple regression model.

Shrinkage Methods

- Model selection may be done using several ways like using model selection criteria such as AIC, BIC, Adjusted R^2 .
- Another way is: fit a model containing all p predictors using a technique that *constrains* or *regularizes* the coefficient estimates, or, shrinks the coefficient estimates towards zero.
- It turns out that shrinking the coefficient estimates can significantly reduce their variance.

Ridge Regression - L2 Regularization

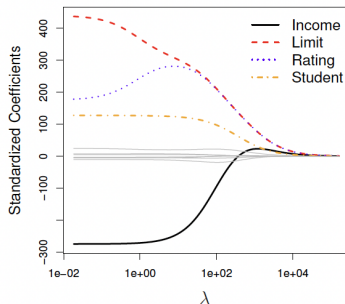
- Recall: The least squares (LS) fitting procedure estimates $\beta_0, \beta_1, \dots, \beta_p$ using values that minimize:
$$SSE = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2.$$
- *Ridge regression* is similar to LS, except that the coefficients are estimated by minimizing a slightly different quantity.
- The ridge regression coefficient estimates $\hat{\beta}^R$ are the values that minimize:
$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2 = SSE + \lambda \sum_{j=1}^p \beta_j^2$$
- Here $\lambda \geq 0$ is a **tuning parameter**, to be determined separately.
- The above has two different parts.
- As with LS, ridge regression seeks coefficient estimates that fit the data well, by making the SSE small.
- The second term, $\lambda \sum_{j=1}^p \beta_j^2$, called a *shrinkage penalty*, is small when $\beta_1, \dots, \beta_p$ are close to 0, and so it has the effect of shrinking the estimates of β_j towards 0.

Ridge Regression - L2 Regularization

- The tuning parameter λ serves to control the relative impact of these 2 terms on the regression coefficient estimates.
- When $\lambda = 0$, the penalty term has no effect, and ridge regression will produce the least squares estimates.
- As $\lambda \rightarrow \infty$, the impact of the shrinkage penalty grows, and the ridge regression coefficient estimates will approach 0.
- Unlike LS, which generates only 1 set of coefficient estimates, ridge regression produces a different set of coefficient estimates, $\hat{\beta}_\lambda^R$, for each value of λ .
- Selecting a good value for λ is critical.
- Note that the shrinkage penalty is applied to $\beta_1, \dots, \beta_p$, but not to the intercept β_0 .
- We want to shrink the estimated association of each variable with the response.

Ridge Regression - L2 Regularization

- However, we do not want to shrink the intercept, which is a measure of the mean value of the response when $x_{i1} = x_{i2} = \dots = x_{ip} = 0$.
- Application to Credit Data (ISLR2).
- Information on credit balance, along with 10 predictors like income, credit limit, credit rating etc. for 400 individuals.



The standardized ridge regression coefficients for the Credit data, as a function of λ .

Ridge Regression - L2 Regularization

- In the figure, each curve corresponds to the ridge regression coefficient estimate for 1 of the 10 variables, plotted as a function of λ .
- At the extreme left-hand side of the plot, λ is 0, so the corresponding ridge coefficient estimates are the same as the usual LS estimates.
- As λ increases, the ridge coefficient estimates shrink towards 0.
- When λ is extremely large, all of the ridge coefficient estimates are basically 0; this corresponds to the *null* model with no predictors.
- In this plot, the income, limit, rating, and student variables are displayed in distinct colors, since these variables tend to have the largest coefficient estimates.
- While the ridge coefficient estimates tend to decrease in aggregate as λ increases, individual coefficients may occasionally increase as λ increases.

Standardized Ridge Regression

- The standard LS coefficient estimates are *scale equivariant*.
- Multiplying X_j by a constant c leads to a scaling of the LS coefficient estimates by a factor of $\frac{1}{c}$.
- Regardless of how the j th predictor is scaled, $X_j\hat{\beta}_j$ remains same.
- In contrast, ridge regression coefficient estimates can change substantially when multiplying a given predictor by a constant.
- $X_j\hat{\beta}_{j,\lambda}^R$ will depend not only on the value of λ , but also on the scaling of the j th predictor (even other predictors!)
- Therefore, it is best to apply ridge regression after *standardizing the predictors*, using the formula
$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}},$$
 so that they are all on the *same scale*.
- As a result the final fit will not depend on the scale on which the predictors are measured.

The Lasso - (Least Absolute Shrinkage and Selection Operator) or L1 Regularization

- Ridge regression has one obvious disadvantage - will include all p predictors in the final model.
- The penalty $\lambda \sum \beta_j^2$ will shrink all of the coefficients towards 0, but not set any of them exactly to 0 (unless $\lambda = \infty$).
- This may not be a problem for prediction accuracy, but can be a challenge in model interpretation when the number of variables p is quite large.
- The Lasso is a relatively recent alternative to ridge regression that overcomes this disadvantage.
- The Lasso coefficients, $\hat{\beta}_\lambda^L$, minimize:
$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j|$$
$$= SSE + \lambda \sum_{j=1}^p |\beta_j|$$
- The Lasso shrinks the coefficient estimates towards 0 and forces some coefficient estimates to be exactly 0.
- Thus, the Lasso performs *variable selection*.

The Lasso vs. Ridge Regression

- Neither ridge regression nor the Lasso will universally dominate the other.
- Expect the Lasso to perform better when a relatively small number of predictors have substantial coefficients, and remaining predictors have coefficients that are very small or zero.
- Ridge regression will perform better when the response is a function of many predictors, all with coefficients of roughly equal size.

Dimension Reduction: Principal Components Regression

- Principal components analysis (PCA) is a popular approach for deriving a low-dimensional set of features from a large set of variables.
- Technique to reduce dimension of an $n \times p$ data matrix $\mathbf{X}$, when p is large.
- We wish to visualize n observations with measurements on a set of p features, $X_1, X_2, \dots, X_p$.
- PCA finds a low-dimensional representation of the data set that contains as much as possible of the variation.
- We seek a small number of dimensions to explain maximum variation in the data, where each dimension is a linear combination of the p features.

Principal Components

- The *first principal component* of a set of features $X_1, X_2, \dots, X_p$ is a linear combination of the features that has the largest variance:
$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p.$$
- This is subject to $\sum_{j=1}^p \phi_{j1}^2 = 1$.
- We refer to the elements $\phi_{11}, \phi_{21}, \dots, \phi_{p1}$ as the loadings of the first principal loading component.
- Together, the loadings make up the *principal component loading vector*, $\phi_1 = (\phi_{11}, \phi_{21}, \dots, \phi_{p1})^T$.
- We obtain these by solving an optimization problem.

Principal Components

- After the first principal component Z_1 of the features has been determined, we can find the second principal component Z_2 .
- Z_2 is the linear combination of $X_1, X_2, \dots, X_p$ that has maximal variance out of all linear combinations that are uncorrelated with Z_1 .
- Thus, the transformed predictors $Z_1, Z_2, \dots, Z_M$ are obtained.
- Note that $M \ll p$.
- The model is fit using these M predictors.

An Introduction to Statistical Learning by G.James, D.Witten,
T.Hastie, R.Tibshirani